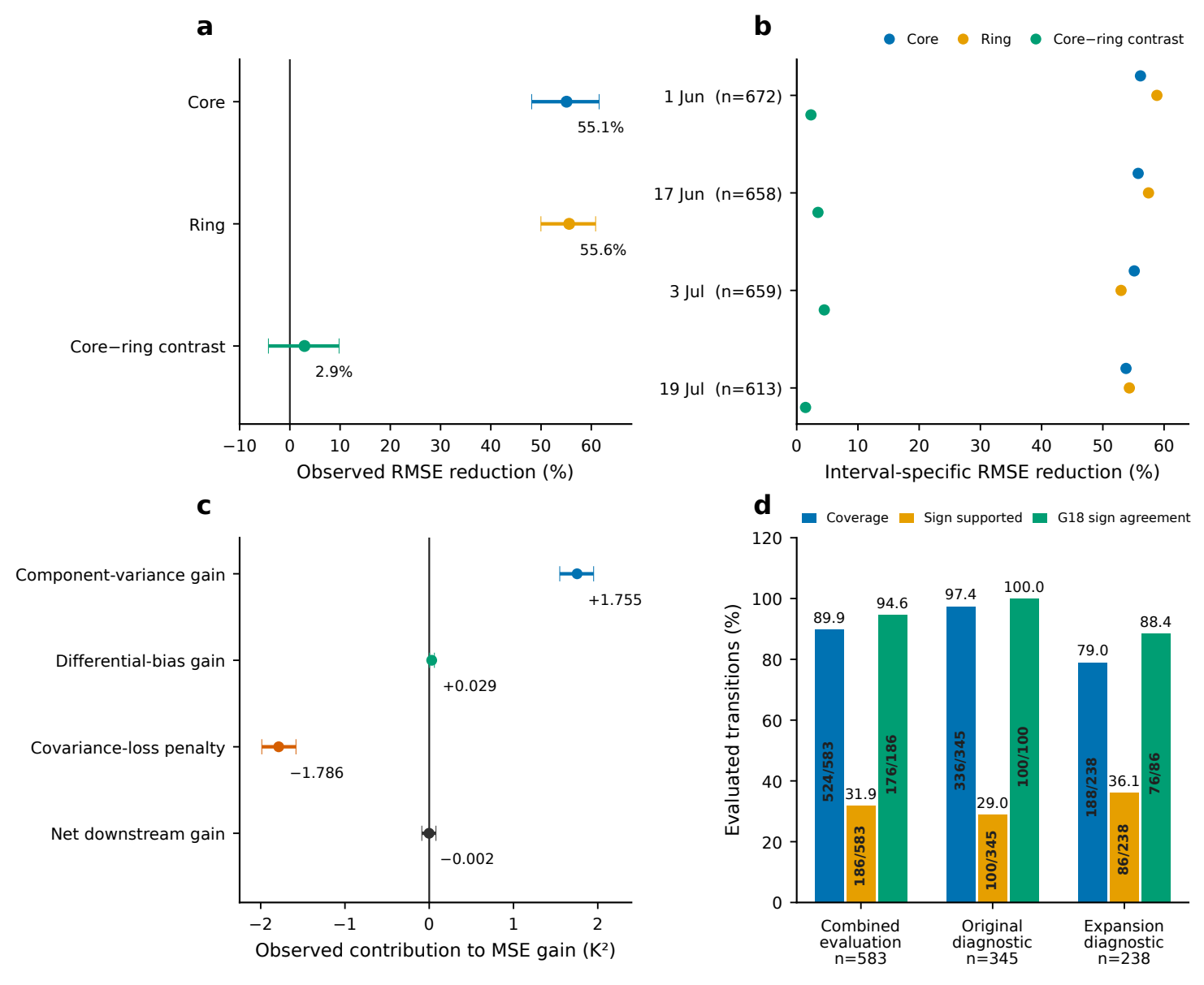
a
Core
Ring
Core−ring contrast
55.1%
55.6%
2.9%
−10
0
10
20
30
40
50
60
Observed RMSE reduction (%)
b
Core
Ring
Core−ring contrast
1 Jun (n=672)
17 Jun (n=658)
3 Jul (n=659)
19 Jul (n=613)
0
10
20
30
40
50
60
Interval-specific RMSE reduction (%)
c
Component-variance gain
Differential-bias gain
Covariance-loss penalty
Net downstream gain
+1.755
+0.029
−1.786
−0.002
−2
−1
0
1
2
Observed contribution to MSE gain (K²)
d
Coverage
Sign supported
G18 sign agreement
Evaluated transitions (%)
0
20
40
60
80
100
120
89.9
31.9
94.6
524/583
186/583
176/186
97.4
29.0
100.0
336/345
100/345
100/100
79.0
36.1
88.4
188/238
86/238
76/86
Combined evaluation n=583
Original diagnostic n=345
Expansion diagnostic n=238